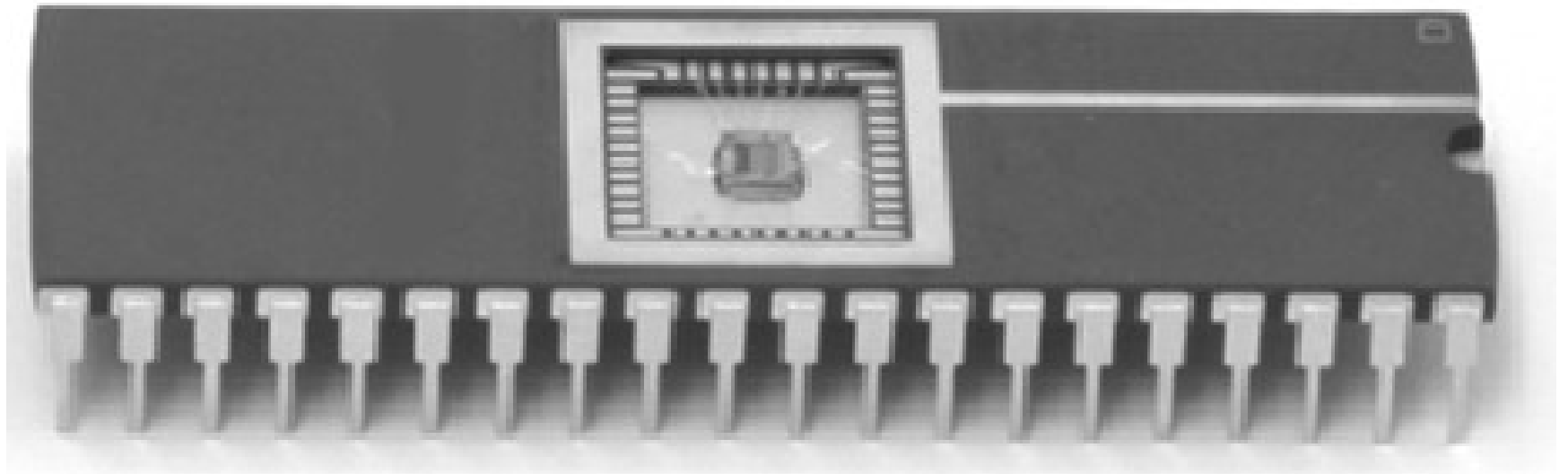


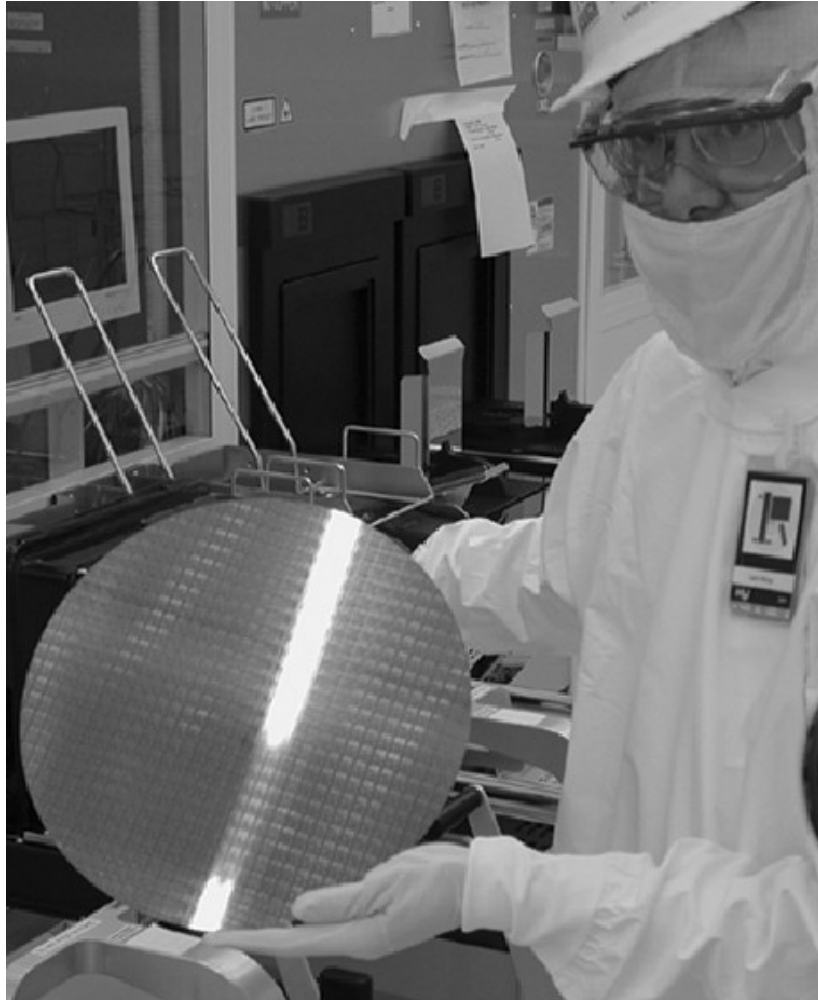
Técnicas Digitales I

Luis Eduardo Toledo

CIRCUITO INTEGRADO



OBLEA DE SILICIO



Logic Levels

- Discrete voltages represent 1 and 0
- For example:
 - 0 = *ground* (GND) or 0 volts
 - 1 = V_{DD} or 5 volts
- What about 4.99 volts? Is that a 0 or a 1?
- What about 3.2 volts?

Logic Levels

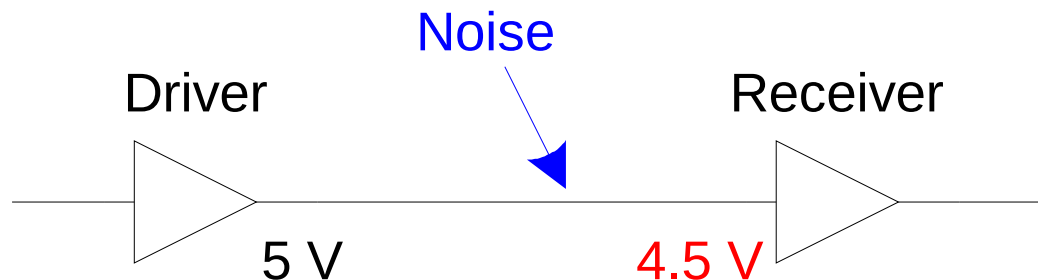
- *Range* of voltages for 1 and 0
- Different ranges for inputs and outputs to allow for *noise*



What is Noise?

What is Noise?

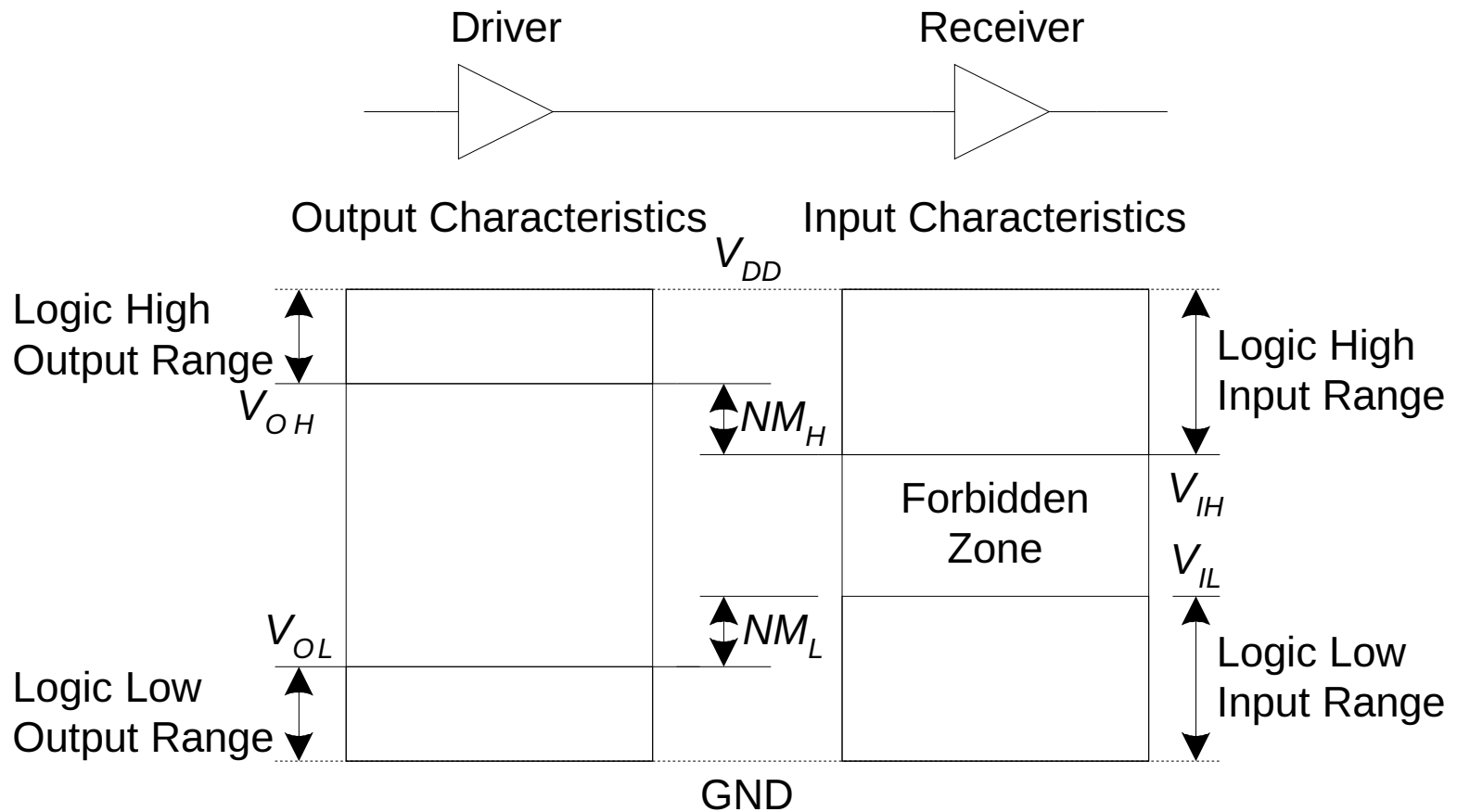
- **Anything that degrades the signal**
 - E.g., resistance, power supply noise, coupling to neighboring wires, etc.
- **Example:** a gate (driver) outputs 5 V but, because of resistance in a long wire, receiver gets 4.5 V



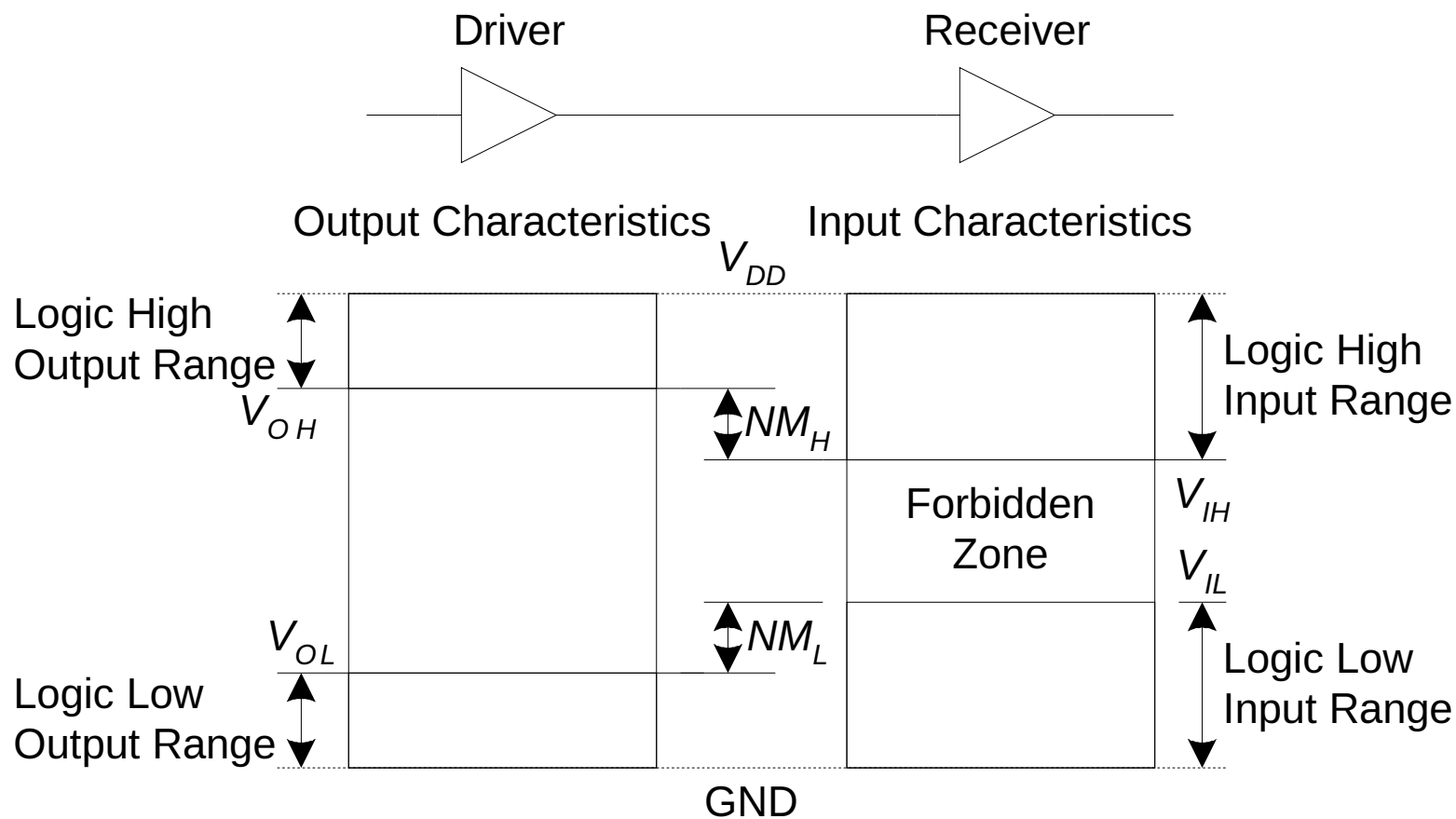
The Static Discipline

- With logically valid inputs, every circuit element must produce logically valid outputs
- Use limited ranges of voltages to represent discrete values

Logic Levels



Noise Margins

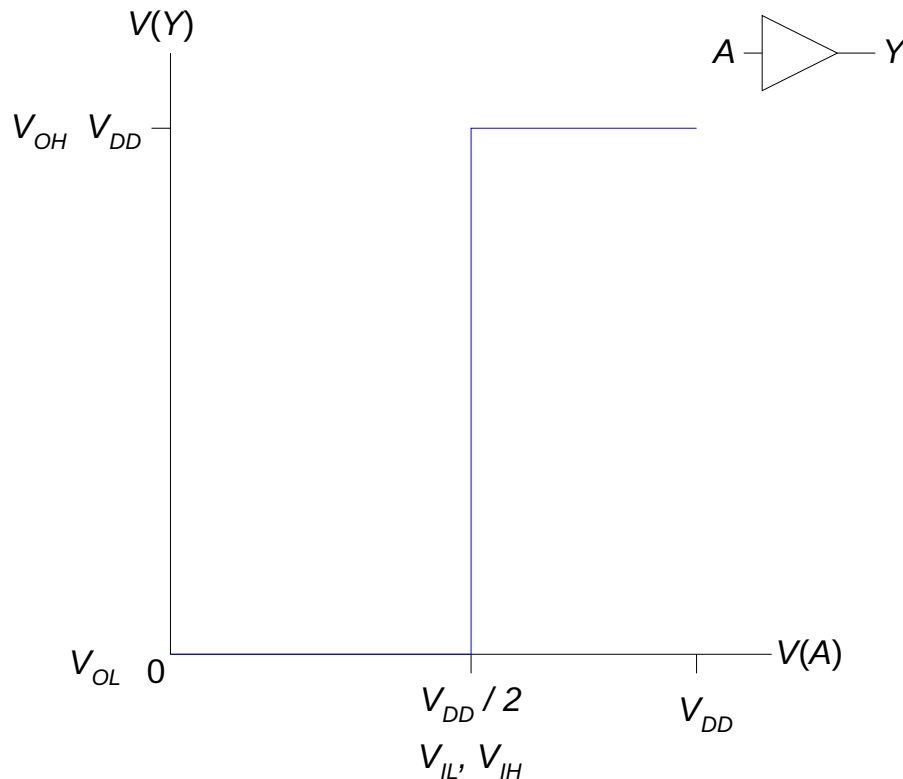


$$NM_H = V_{OH} - V_{IH}$$

$$NM_L = V_{IL} - V_{OL}$$

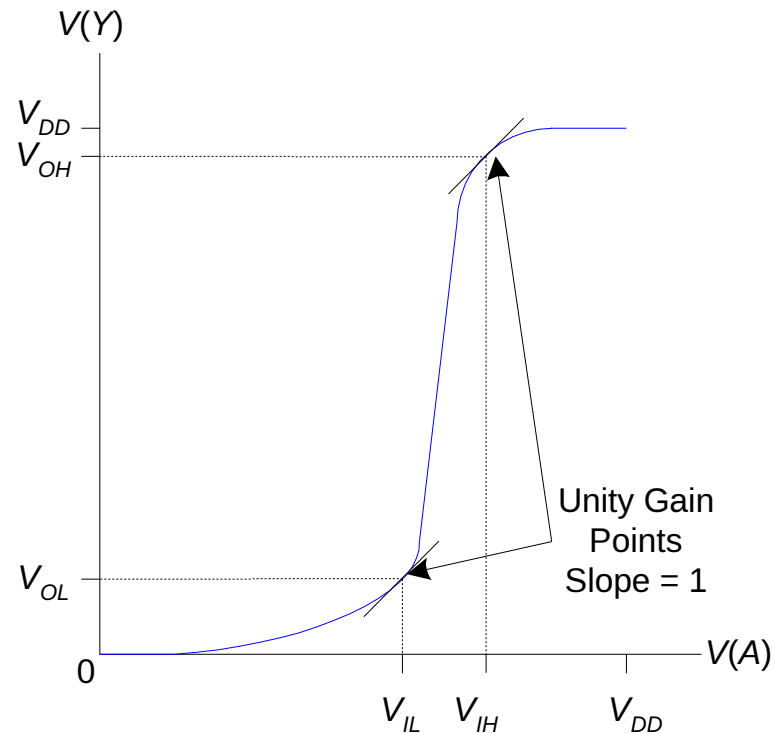
DC Transfer Characteristics

Ideal Buffer:



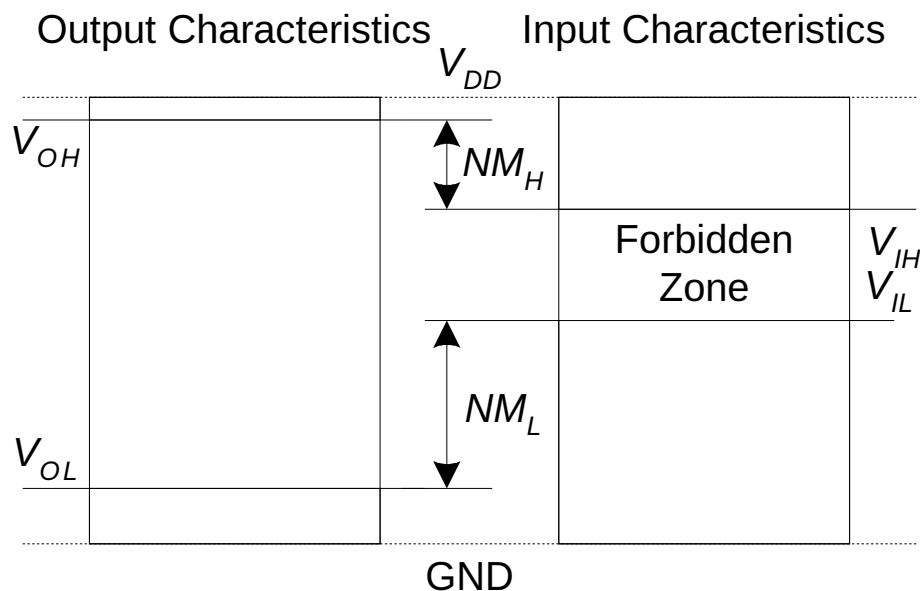
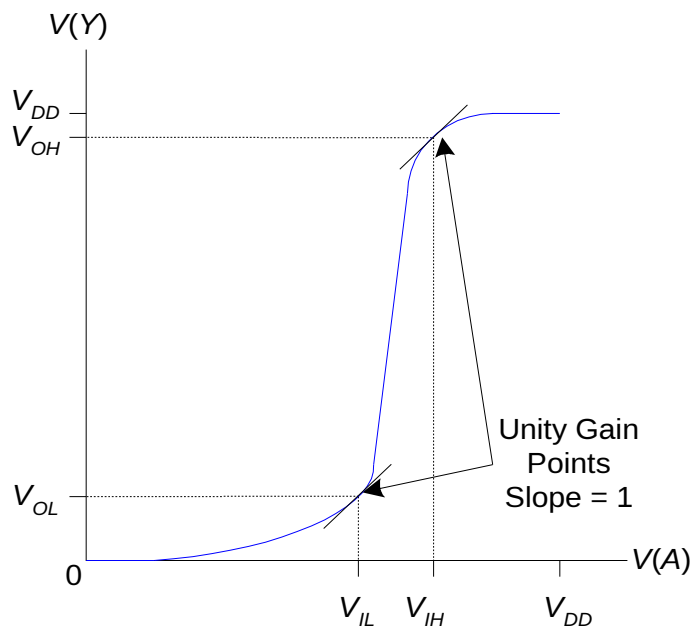
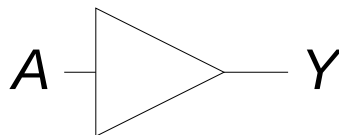
$$NM_H = NM_L = V_{DD}/2$$

Real Buffer:

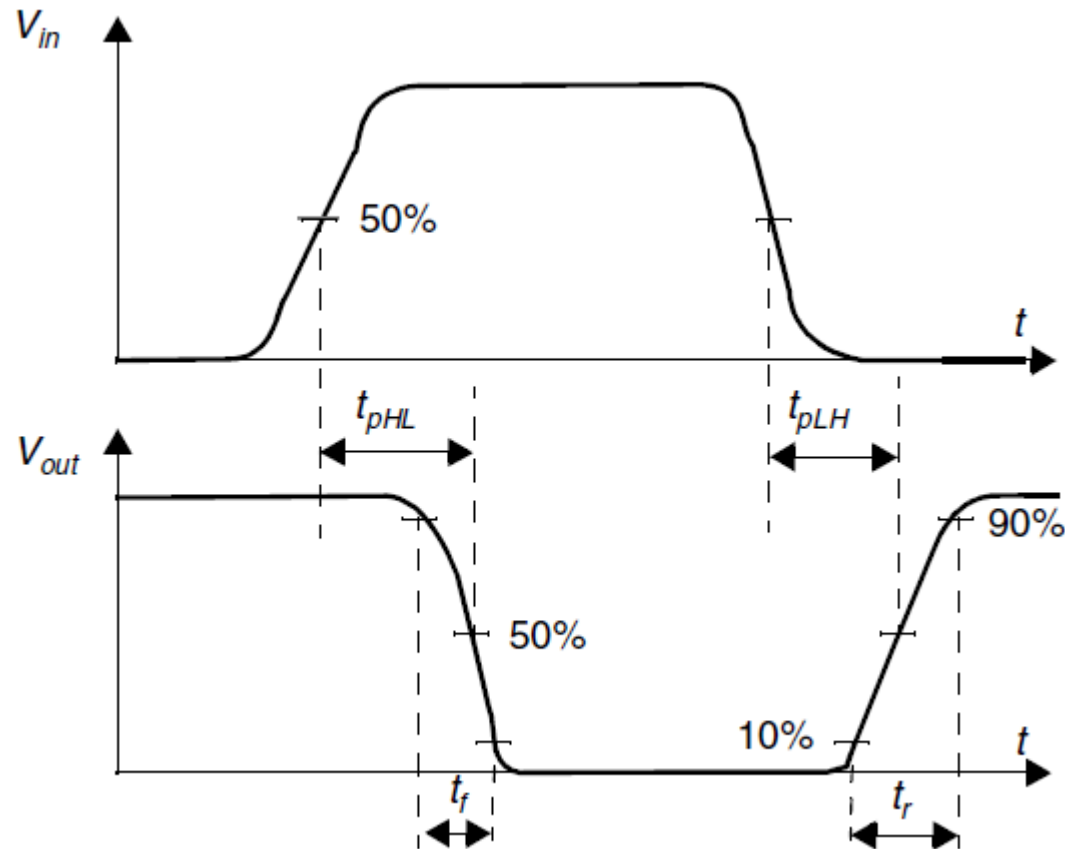


$$NM_H, NM_L < V_{DD}/2$$

DC Transfer Characteristics



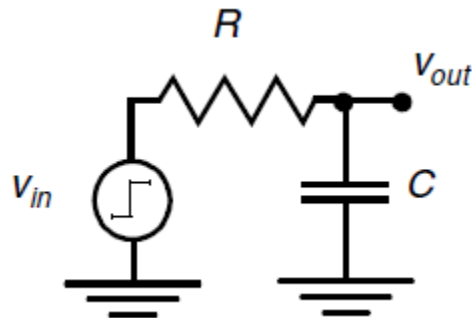
DEFINICIÓN DE RETARDO DE PROPAGACIÓN



$$t_p = \frac{t_{pLH} + t_{pHL}}{2}$$

RETARDO DE PROPAGACIÓN DE UN CIRCUITO RC DE PRIMER ORDEN

LOS CIRCUITOS DIGITALES SE PUEDEN
MODELAR COMO SIMPLES CIRCUITOS RC

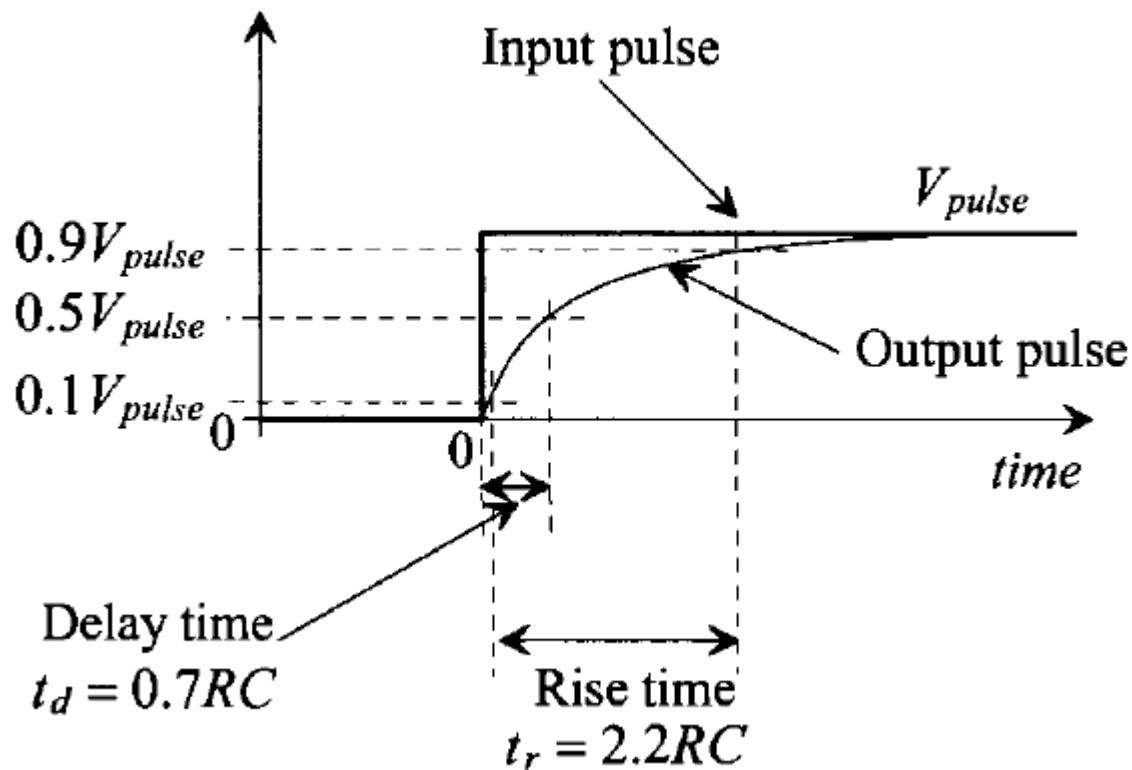


CUANDO SE APLICA UN ESCALON A V_{in}
(DE 0 A V), LA RESPUESTA TRANSITORIA
ES:

$$v_{out}(t) = (1 - e^{-t/\tau}) V$$

$$\tau = RC$$

RETARDO DE PROPAGACIÓN DE UN CIRCUITO RC DE PRIMER ORDEN



$$\frac{V_{pulse}}{2} = V_{pulse}(1 - e^{-t_d/RC}) \rightarrow t_d \approx 0.7RC$$

$$0.1V_{pulse} = V_{pulse}(1 - e^{-t_{10\%}/RC})$$

$$0.9V_{pulse} = V_{pulse}(1 - e^{-t_{90\%}/RC})$$

$$t_r = t_{90\%} - t_{10\%} \approx 2.2RC$$

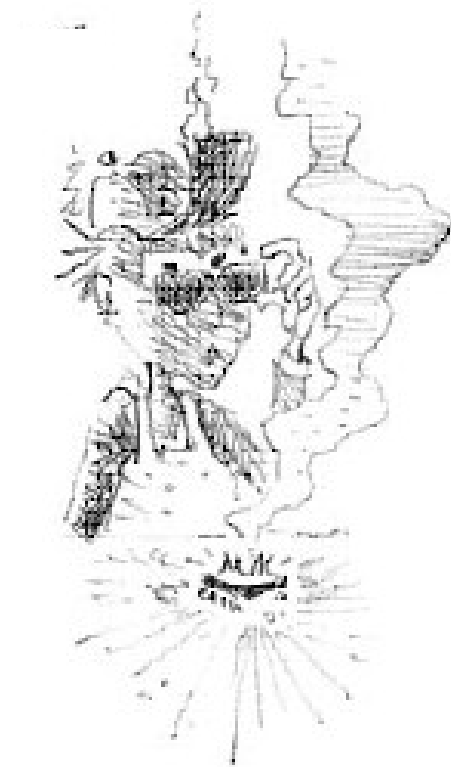
V_{DD} Scaling

- In 1970's and 1980's, $V_{DD} = 5\text{ V}$
- V_{DD} has dropped
 - Avoid frying tiny transistors
 - Save power
- 3.3 V, 2.5 V, 1.8 V, 1.5 V, 1.2 V, 1.0 V, ...
- Be careful connecting chips with different supply voltages

Chips operate because they contain magic smoke

Proof:

- if the magic smoke is let out, the chip stops working

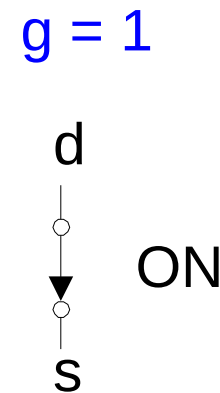
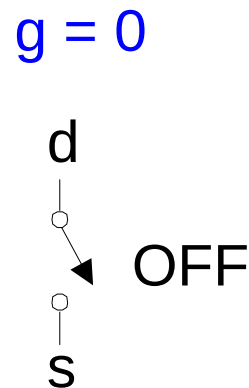
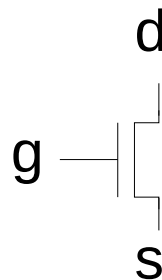


Logic Family Examples

Logic Family	V_{DD}	V_{IL}	V_{IH}	V_{OL}	V_{OH}
TTL	5 (4.75 - 5.25)	0.8	2.0	0.4	2.4
CMOS	5 (4.5 - 6)	1.35	3.15	0.33	3.84
LVTTL	3.3 (3 - 3.6)	0.8	2.0	0.4	2.4
LVC MOS	3.3 (3 - 3.6)	0.9	1.8	0.36	2.7

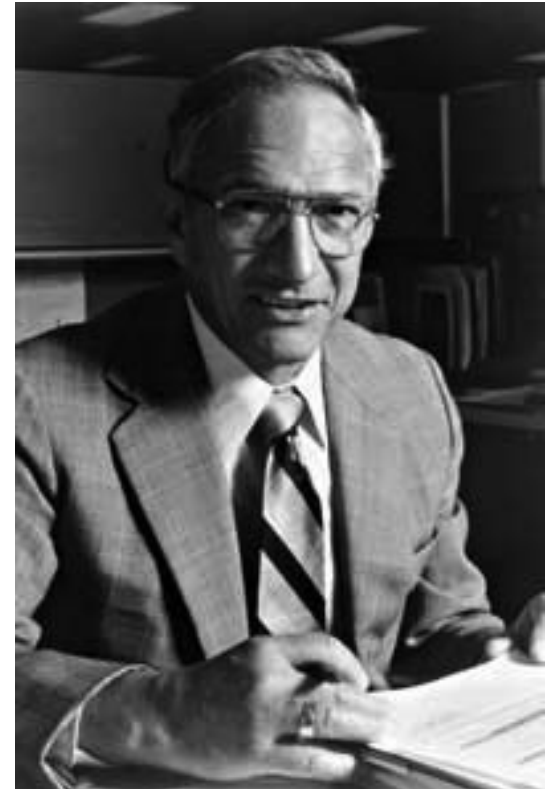
Transistors

- Logic gates built from transistors
- 3-ported voltage-controlled switch
 - 2 ports connected depending on voltage of 3rd
 - d and s are connected (ON) when g is 1



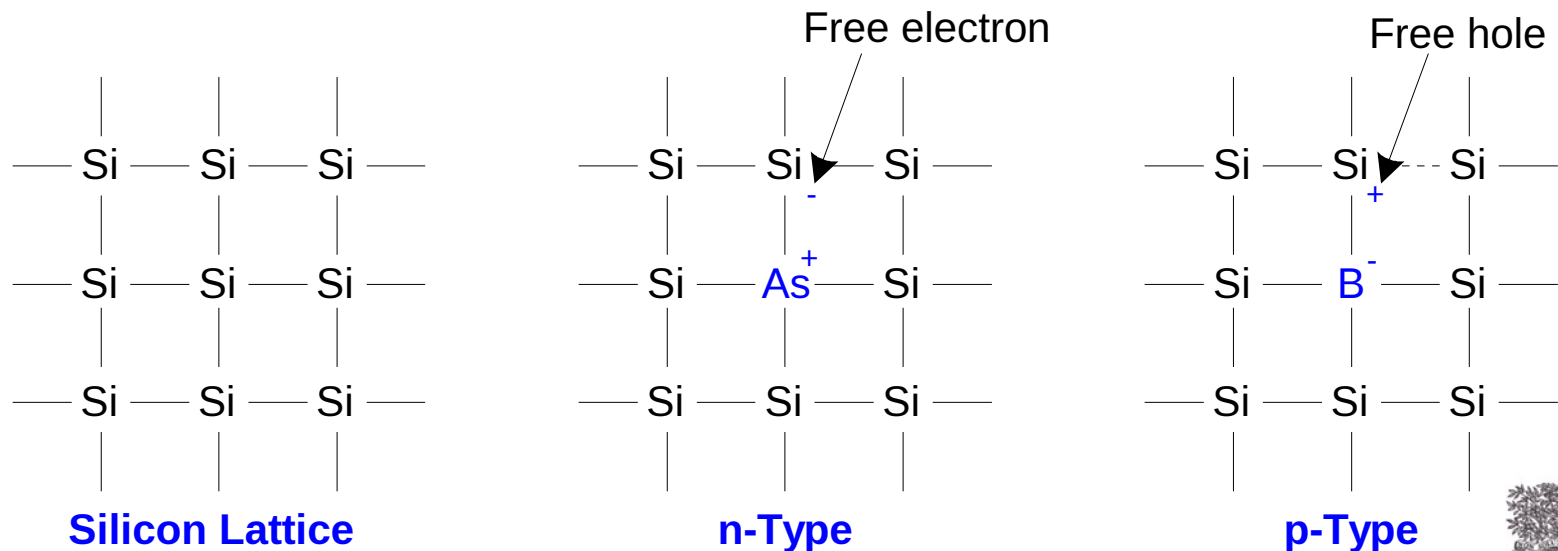
Robert Noyce, 1927-1990

- Nicknamed “Mayor of Silicon Valley”
- Co-founded Fairchild Semiconductor in 1957
- Co-founded Intel in 1968
- Co-invented the integrated circuit



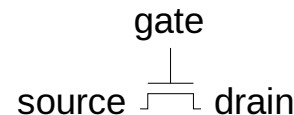
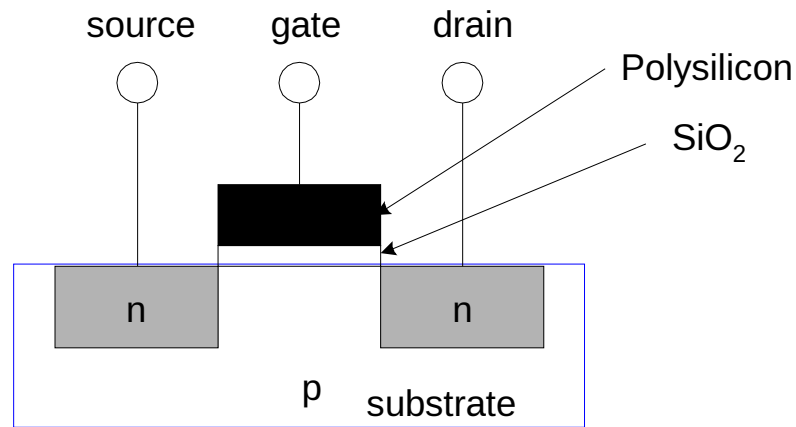
Silicon

- Transistors built from silicon, a semiconductor
- Pure silicon is a poor conductor (no free charges)
- Doped silicon is a good conductor (free charges)
 - n-type (free *negative* charges, electrons)
 - p-type (free *positive* charges, holes)



MOS Transistors

- **Metal oxide silicon (MOS) transistors:**
 - Polysilicon (used to be **metal**) gate
 - **Oxide** (silicon dioxide) insulator
 - Doped **silicon**

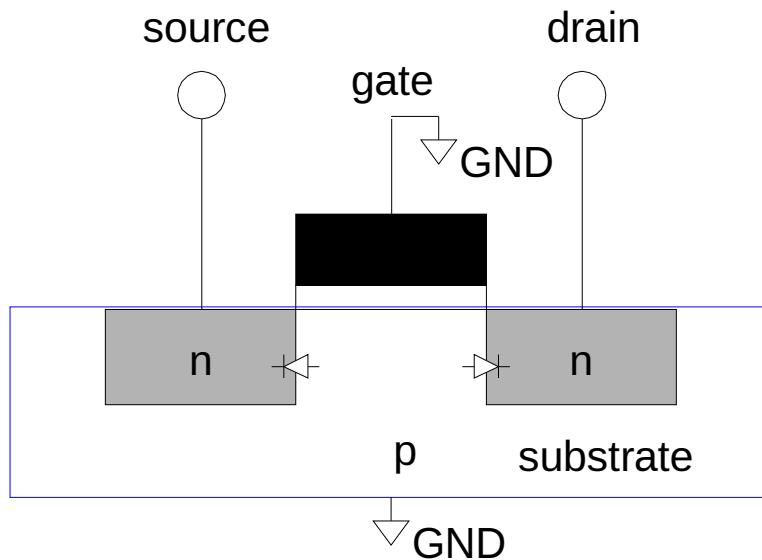


nMOS

Transistors: nMOS

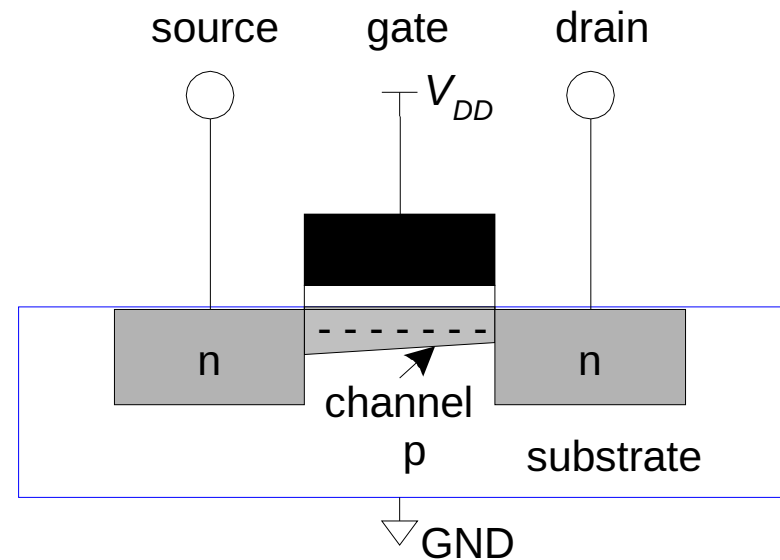
Gate = 0

OFF (no connection between source and drain)



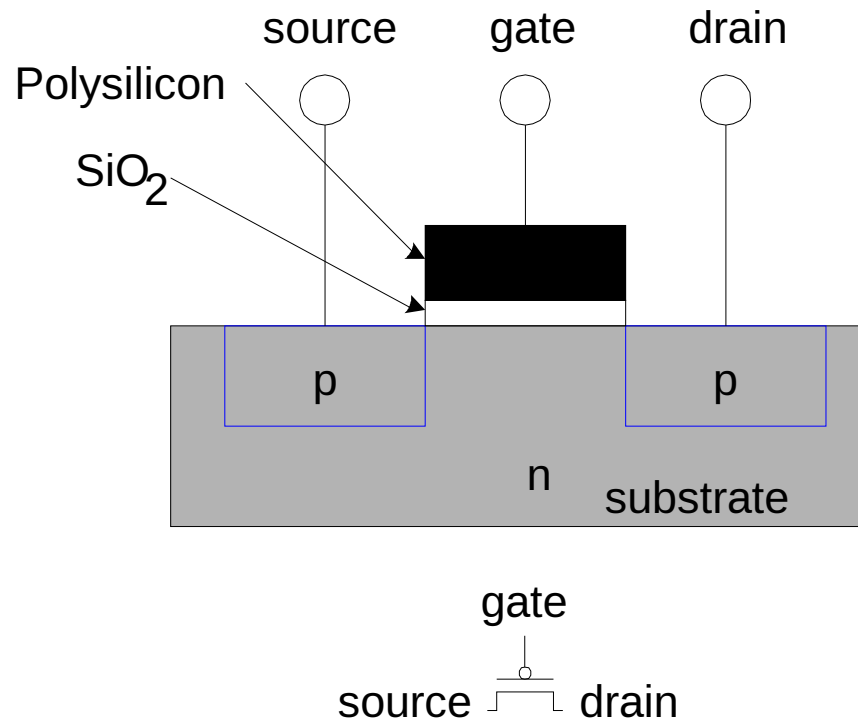
Gate = 1

ON (channel between source and drain)



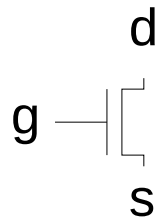
Transistors: pMOS

- pMOS transistor is opposite
 - ON when Gate = 0
 - OFF when Gate = 1

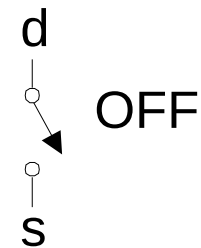


Transistor Function

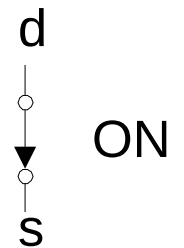
nMOS



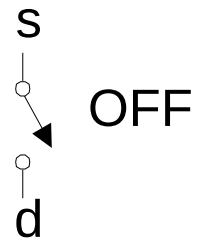
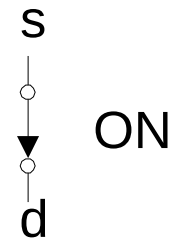
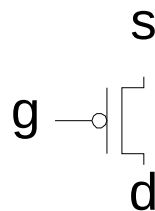
$g = 0$



$g = 1$

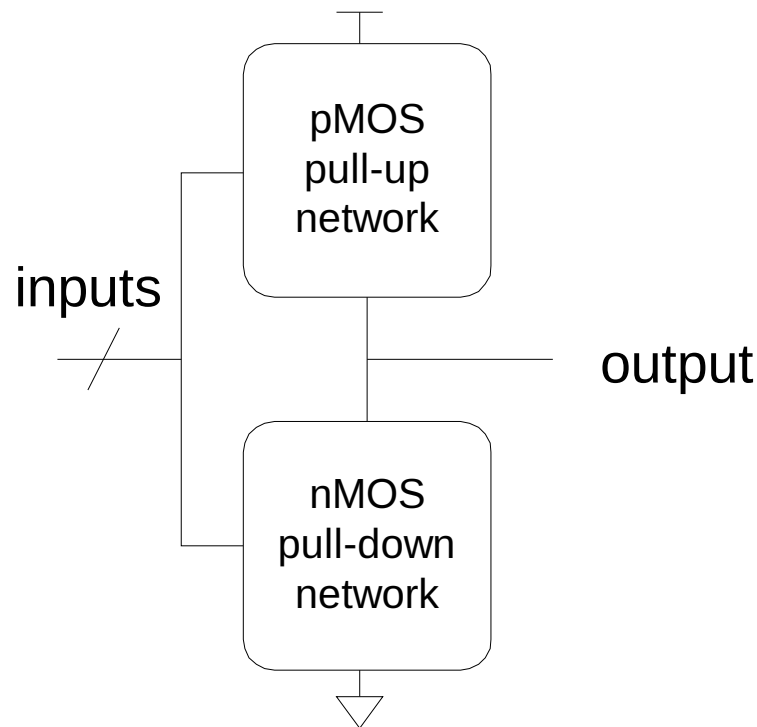


pMOS



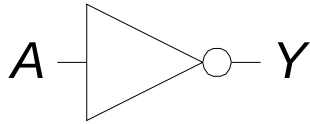
Transistor Function

- **nMOS:** pass good 0's, so connect source to GND
- **pMOS:** pass good 1's, so connect source to V_{DD}



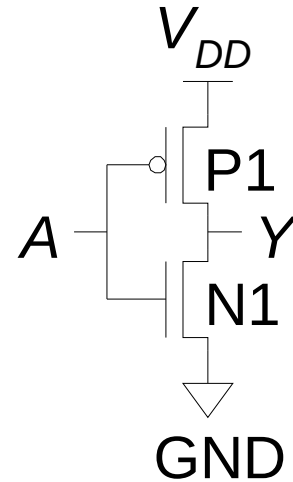
CMOS Gates: NOT Gate

NOT



$$Y = \overline{A}$$

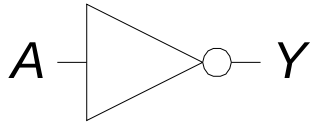
A	Y
0	1
1	0



A	P1	N1	Y
0			
1			

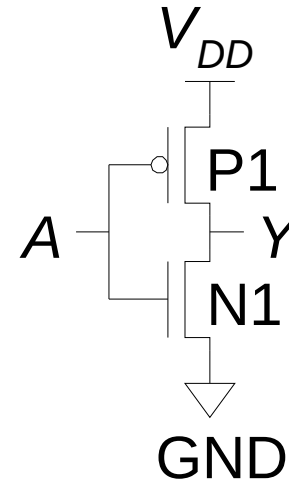
CMOS Gates: NOT Gate

NOT



$$Y = \overline{A}$$

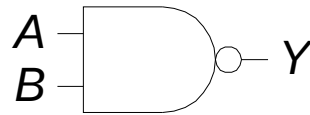
A	Y
0	1
1	0



A	P1	N1	Y
0	ON	OFF	1
1	OFF	ON	0

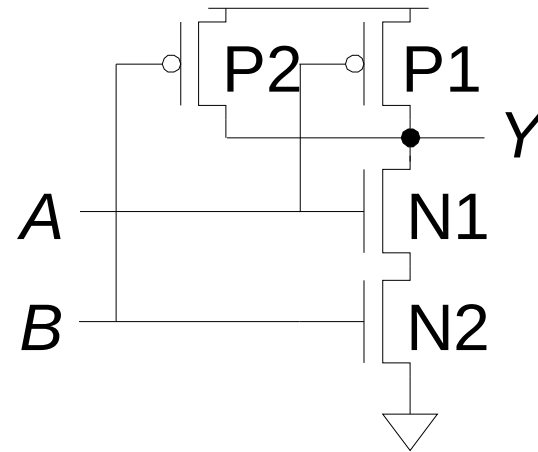
CMOS Gates: NAND Gate

NAND



$$Y = \overline{AB}$$

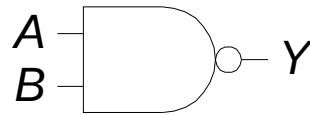
A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0



A	B	P1	P2	N1	N2	Y
0	0					
0	1					
1	0					
1	1					

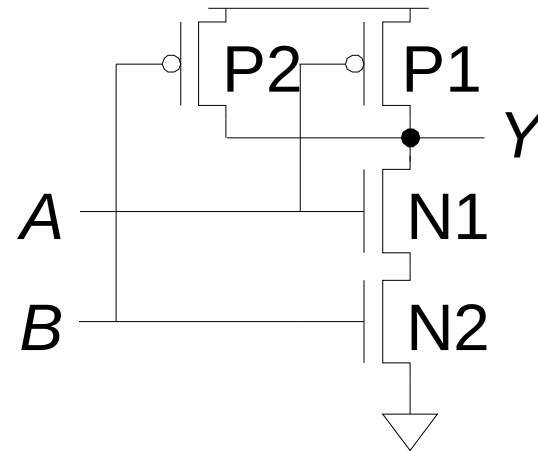
CMOS Gates: NAND Gate

NAND



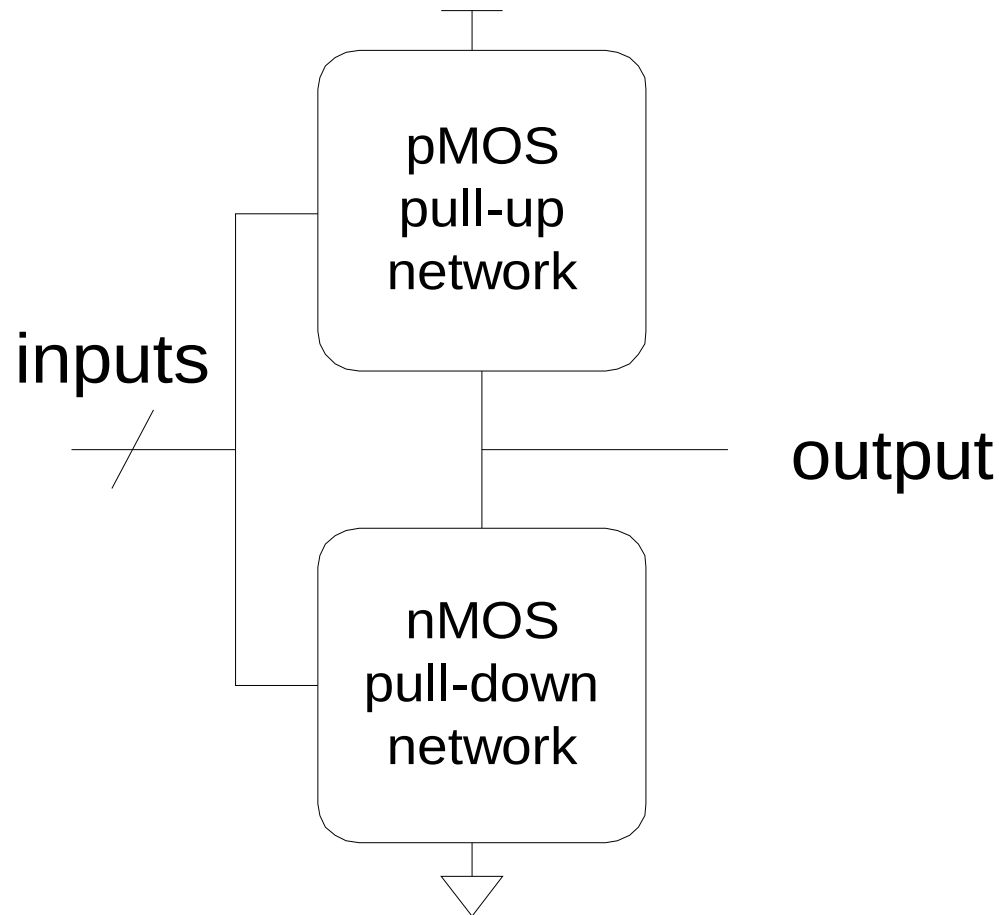
$$Y = \overline{AB}$$

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0



A	B	P1	P2	N1	N2	Y
0	0	ON	ON	OFF	OFF	1
0	1	ON	OFF	OFF	ON	1
1	0	OFF	ON	ON	OFF	1
1	1	OFF	OFF	ON	ON	0

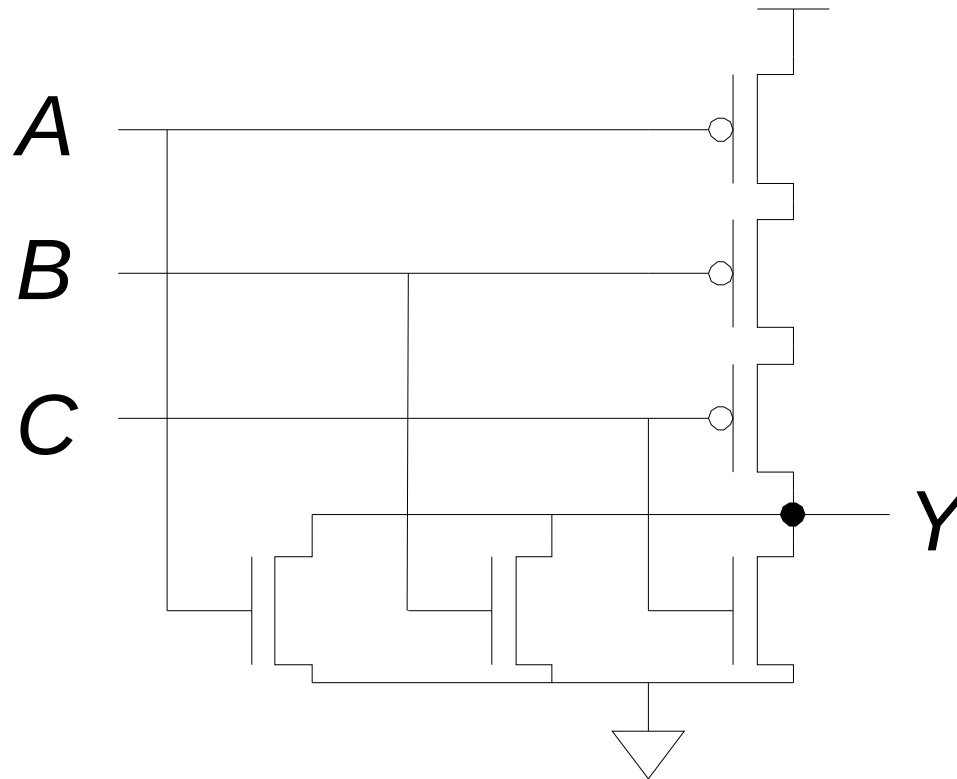
CMOS Gate Structure



NOR Gate

How do you build a three-input NOR gate?

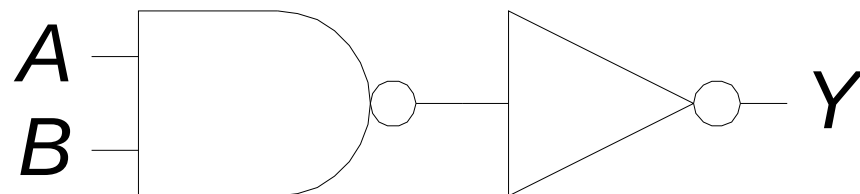
NOR3 Gate



Other CMOS Gates

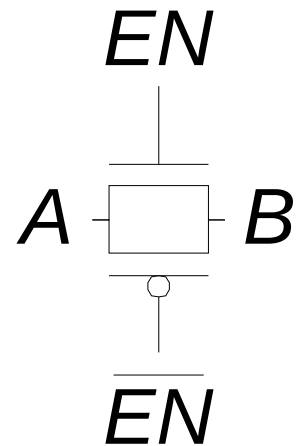
How do you build a two-input AND gate?

AND2 Gate



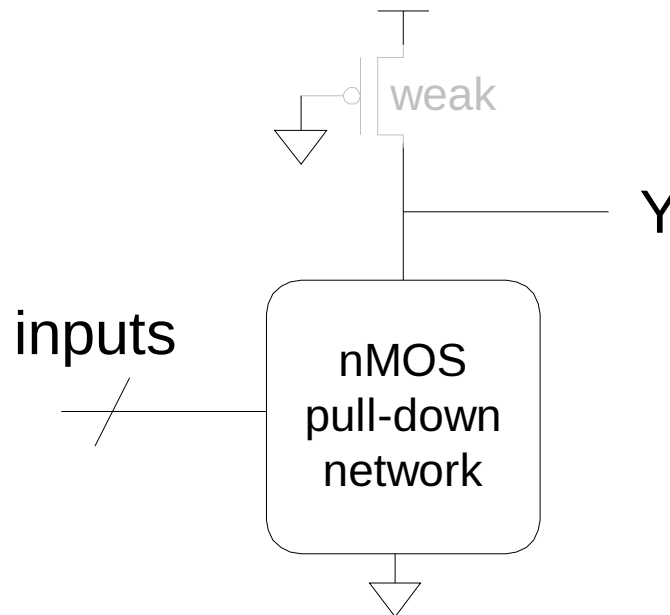
Transmission Gates

- nMOS pass 1's poorly
- pMOS pass 0's poorly
- Transmission gate is a better switch
 - passes both 0 and 1 well
- When $EN = 1$, the switch is ON:
 - $EN = 0$ and A is connected to B
- When $EN = 0$, the switch is OFF:
 - A is not connected to B



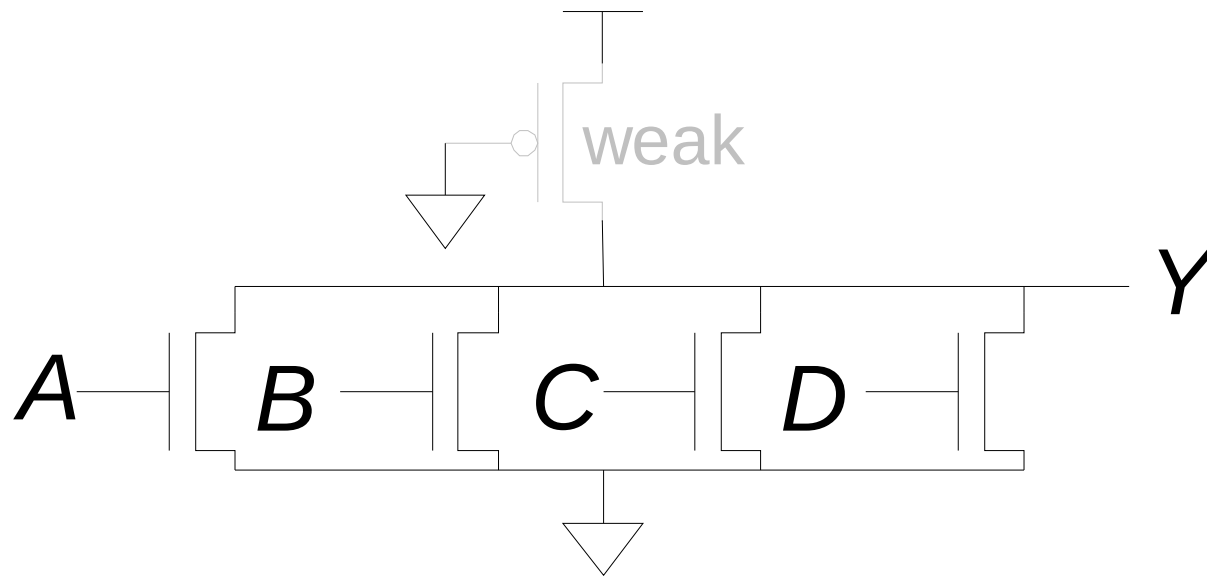
Pseudo-nMOS Gates

- Replace pull-up network with *weak* pMOS transistor that is always on
- pMOS transistor: pulls output HIGH *only* when nMOS network not pulling it LOW



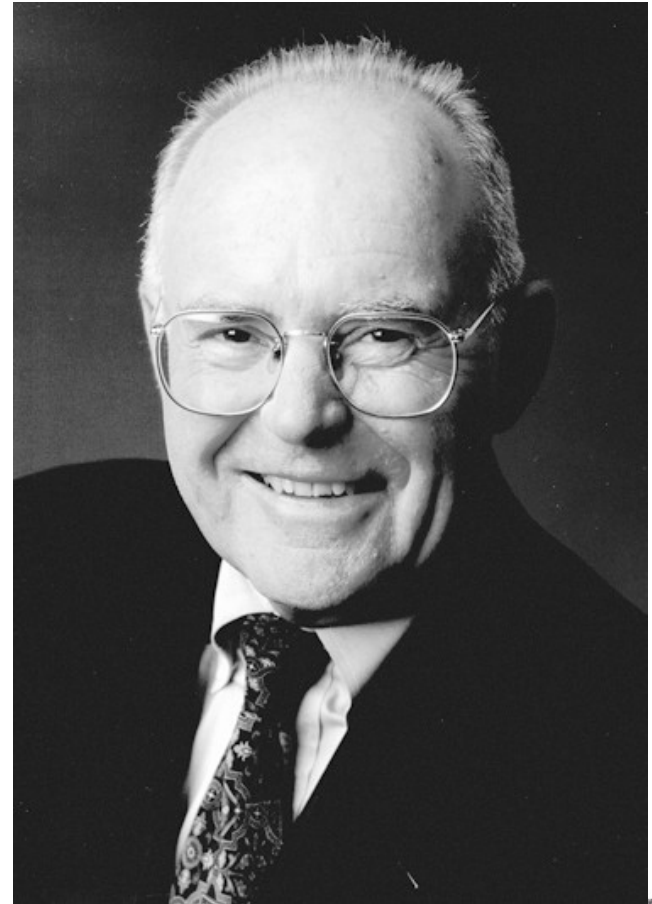
Pseudo-nMOS Example

Pseudo-nMOS NOR4

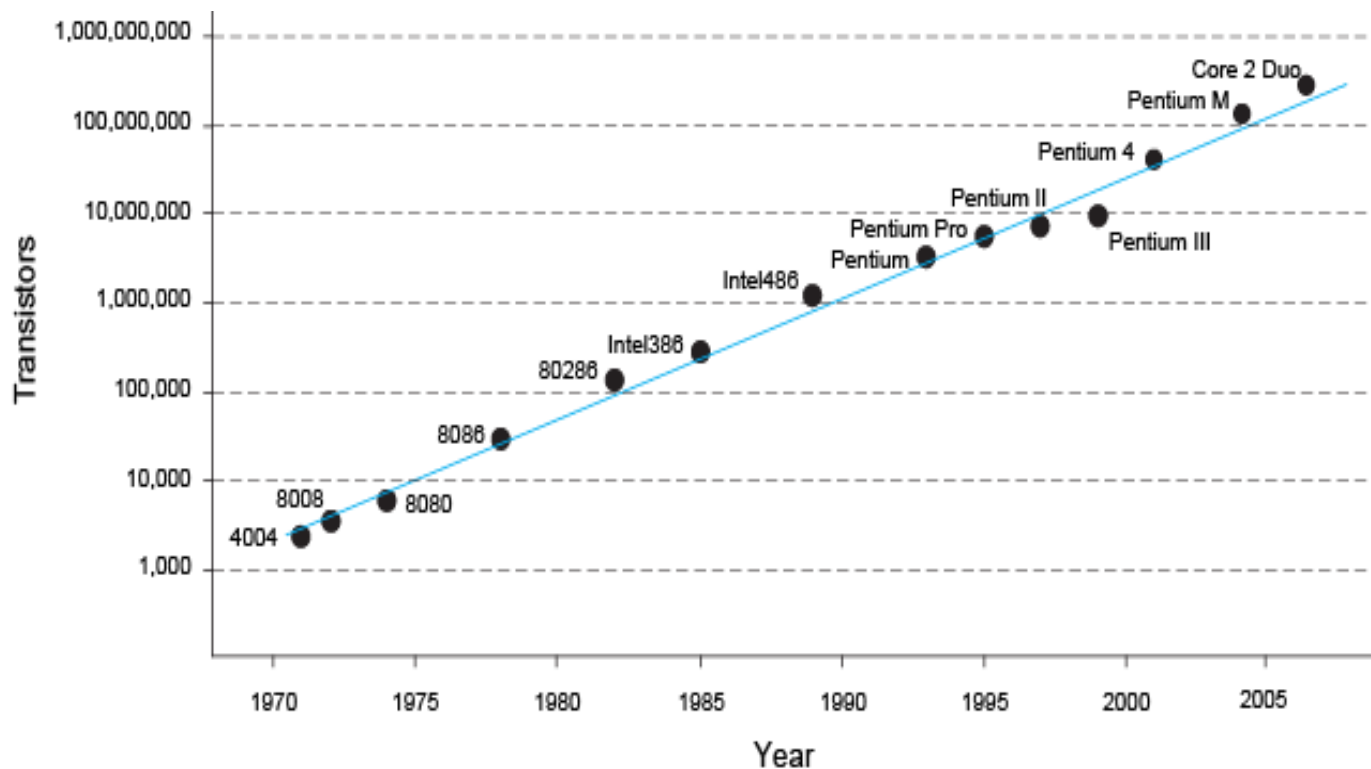


Gordon Moore, 1929-

- Cofounded Intel in 1968 with Robert Noyce.
- **Moore's Law:** number of transistors on a computer chip doubles every year (observed in 1965)
- Since 1975, transistor counts have doubled every two years.



Moore's Law



- “If the automobile had followed the same development cycle as the computer, a Rolls-Royce would today cost \$100, get one million miles to the gallon, and explode once a year . . .”

– Robert Cringley

Power Consumption

- Power = Energy consumed per unit time
 - Dynamic power consumption
 - Static power consumption

Dynamic Power Consumption

- **Power to charge transistor gate capacitances**
 - Energy required to charge a capacitance, C , to V_{DD} is CV_{DD}^2
 - Circuit running at frequency f : transistors switch (from 1 to 0 or vice versa) at that frequency
 - Capacitor is charged $f/2$ times per second (discharging from 1 to 0 is free)
- Dynamic power consumption:

$$P_{dynamic} = \frac{1}{2}CV_{DD}^2f$$

Static Power Consumption

- Power consumed when no gates are switching
- Caused by the *quiescent supply current*, I_{DD} (also called the *leakage current*)
- Static power consumption:

$$P_{static} = I_{DD} V_{DD}$$

Power Consumption Example

- Estimate the power consumption of a wireless handheld computer
 - $V_{DD} = 1.2 \text{ V}$
 - $C = 20 \text{ nF}$
 - $f = 1 \text{ GHz}$
 - $I_{DD} = 20 \text{ mA}$

Power Consumption Example

- Estimate the power consumption of a wireless handheld computer
 - $V_{DD} = 1.2 \text{ V}$
 - $C = 20 \text{ nF}$
 - $f = 1 \text{ GHz}$
 - $I_{DD} = 20 \text{ mA}$

$$\begin{aligned} P &= \frac{1}{2} C V_{DD}^2 f + I_{DD} V_{DD} \\ &= \frac{1}{2} (20 \text{ nF}) (1.2 \text{ V})^2 (1 \text{ GHz}) + \\ &\quad (20 \text{ mA}) (1.2 \text{ V}) \\ &= (14.4 + 0.024) \text{ W} \approx 14.4 \text{ W} \end{aligned}$$

MODELADO A NIVEL DE LLAVES CON VERILOG HDL

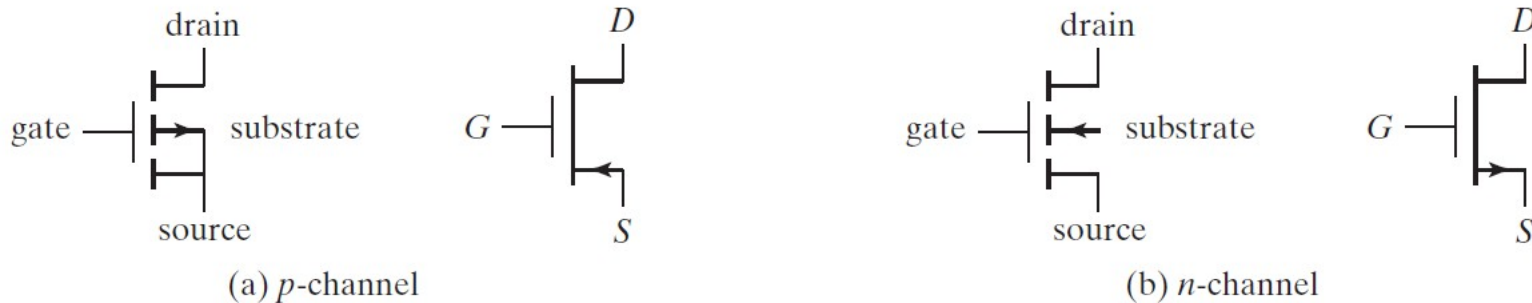


FIGURE A.2

Symbols for MOS transistors

```
nmos (drain, source, gate);  
pmos (drain, source, gate);
```

```
supply1 PWR;  
supply0 GRD;
```

FUNCIÓN NOT

HDL Example A.1

// CMOS inverter of Fig. A.4(a)

module inverter (Y, A);

input A;

output Y;

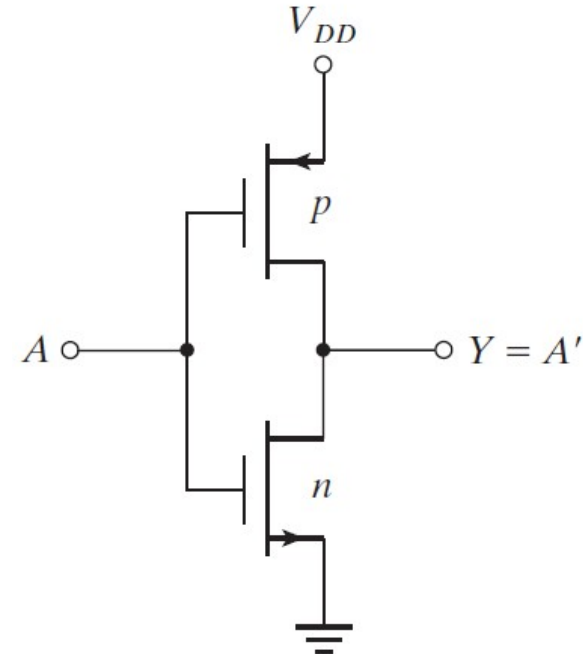
supply1 PWR;

supply0 GRD;

pmos (Y, PWR, A); // (Drain, source, gate)

nmos (Y, GRD, A); // (Drain, source, gate)

endmodule



(a) Inverter

FUNCIÓN NAND

HDL Example A.2

// CMOS two-input NAND of Fig. A.4(b)

```
module NAND2 (Y, A, B);
```

```
  input A, B;
```

```
  output Y;
```

```
  supply1 PWR;
```

```
  supply0 GRD;
```

```
  wire W1;
```

```
  pmos (Y, PWR, A);
```

```
  pmos (Y, PWR, B);
```

```
  nmos (Y, W1, A);
```

```
  nmos (W1, GRD, B);
```

```
endmodule
```

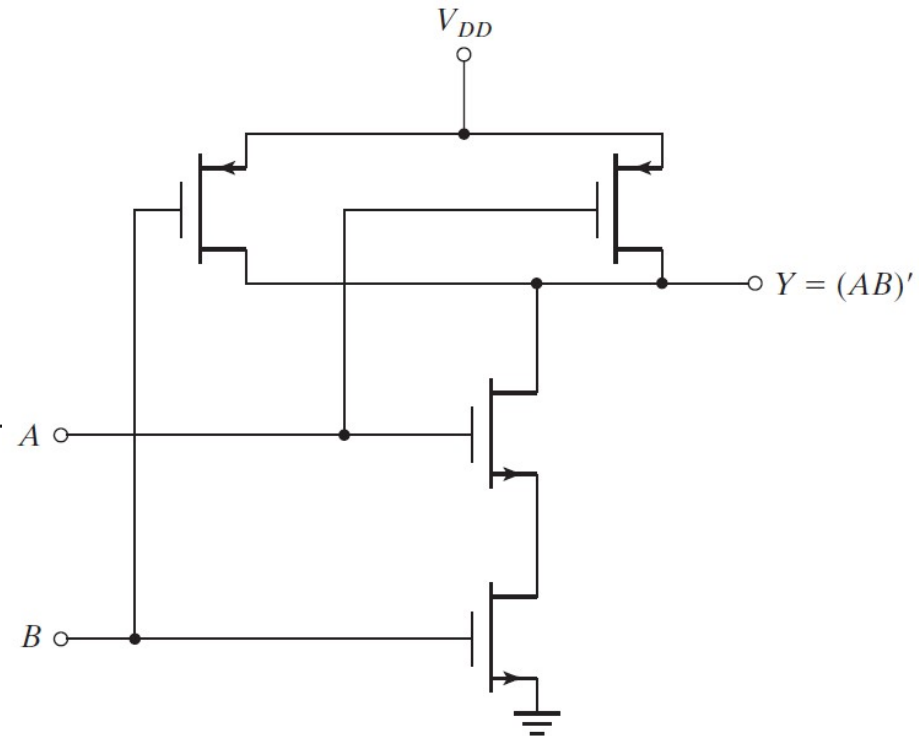
```
  // terminal between two nmos
```

```
  // source connected to Vdd
```

```
  // parallel connection
```

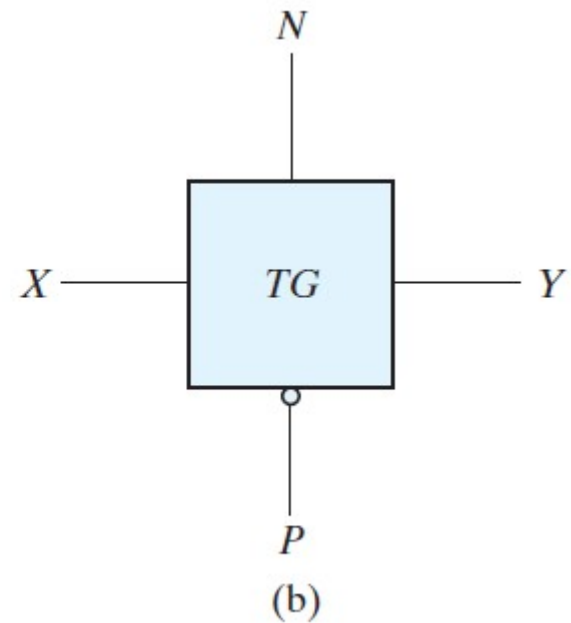
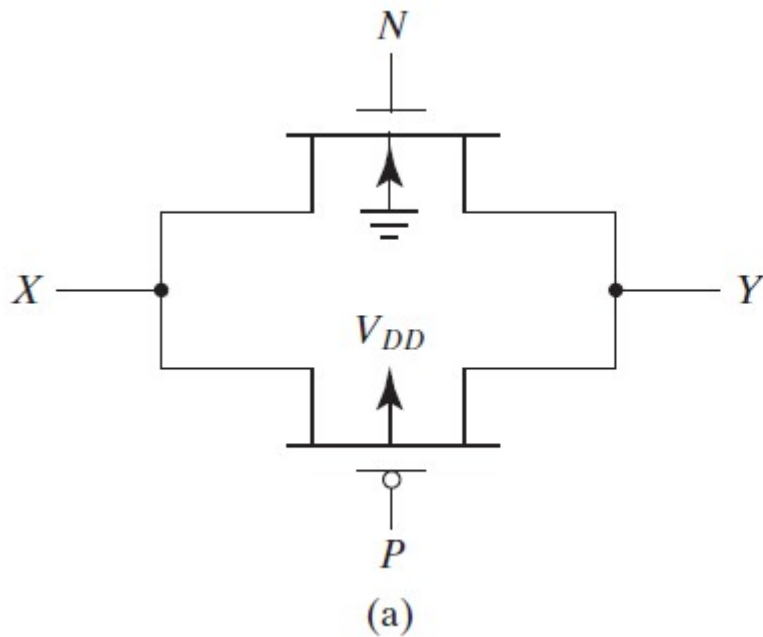
```
  // serial connection
```

```
  // source connected to ground
```



(b) NAND gate

PUERTAS DE TRANSMISIÓN



cmos (output, input, ncontrol, pcontrol); // general description

cmos (Y, X, N, P); // transmission gate of Fig. A.6(b)

FUNCIÓN XOR

//CMOS_XOR with CMOS switches,

module CMOS_XOR (A, B, Y);

input A, B;

output Y;

wire A_b, B_b;

 // instantiate inverter

 inverter v1 (A_b, A);

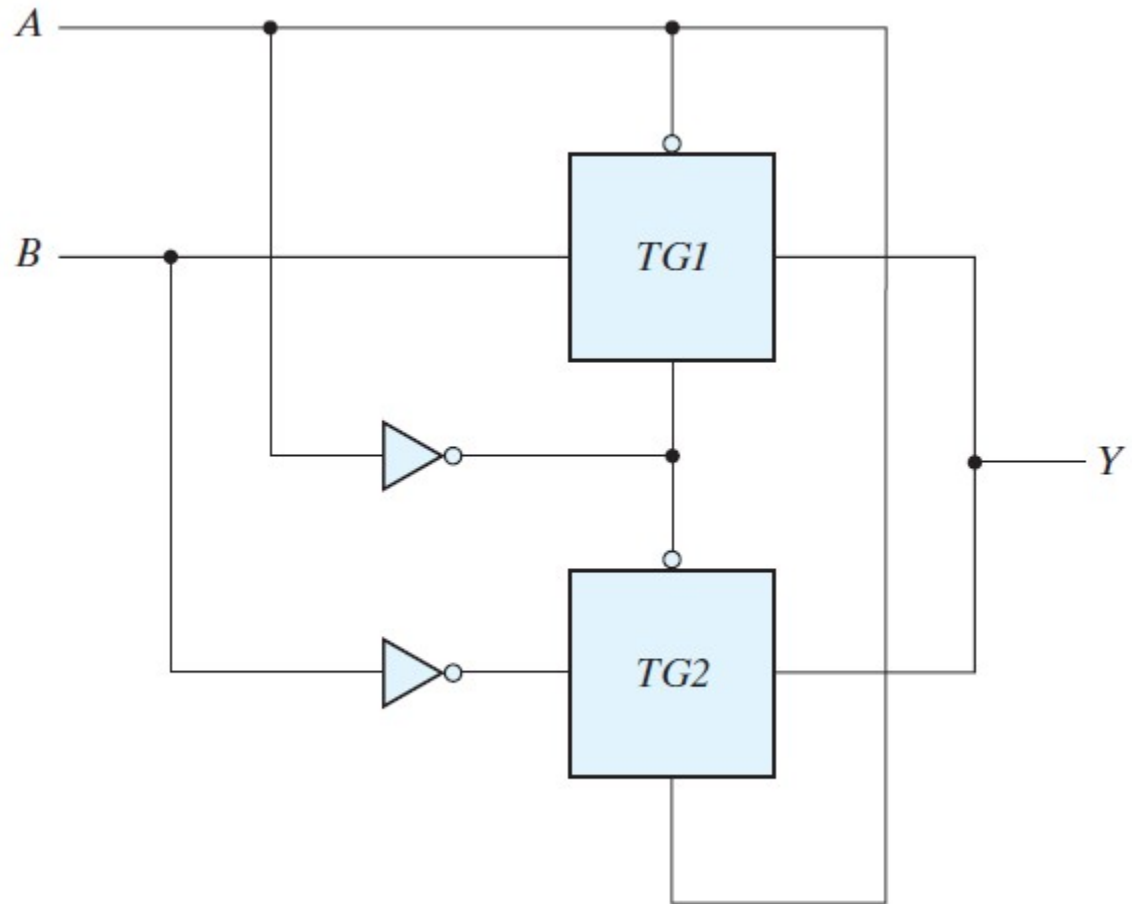
 inverter v2 (B_b, B);

 // instantiate cmos switch

cmos (Y, B, A_b, A);

cmos (Y, B_b, A, A_b);

endmodule



//(output, input, ncontrol, pcontrol)